K - Means Clustering

Instructions:
Please share your answers filled in-line in the word document. Submit code separately wherever applicable.

Please ensure you update all the details:
Name: _____ ROLL NO. _____
Topic: K Means Clustering

Problem Statements:

1. Perform K means clustering on the airlines dataset to obtain optimum number of clusters. Draw the inferences from the clusters obtained. Refer to EastWestAirlines.xlsx dataset.

	ID.	Balance	Qual_miles	cc1_miles	cc2_miles	cc3_miles	Bonus_miles	Bonus_trans	Flight_miles_12mo	Flight_trans_12	Days_since_enroll	Award.
1	1	28143	0	1	1	1	174	1	0	0	7000	0
2	2	19244	0	1	1	1	215	2	0	0	6968	0
3	3	41354	0	1	1	1	4123	4	0	0	7034	0
4	4	14776	0	1	1	1	500	1	0	0	6952	0
5	5	97752	0	4	1	1	43300	26	2077	4	6935	1
6	6	16420	0	1	1	1	0	0	0	0	6942	0
7	7	84914	0	3	1	1	27482	25	0	0	6994	0
8	8	20856	0	1	1	1	5250	4	250	1	6938	1
9	9	443003	0	3	2	1	1753	43	3850	12	6948	1
10	10	104860	0	3	1	1	28426	28	1150	3	6931	1
11	11	40091	0	2	1	1	7278	10	0	0	6959	0
12	12	96522	0	5	1	1	61105	19	0	0	6924	1
13	13	43382	0	2	1	1	11150	20	0	0	6924	0
14	14	43097	0	1	1	1	3258	6	0	0	6918	0
15	15	17648	0	1	1	1	0	0	0	0	6912	0

2. Perform clustering for the crime data and identify the number of clusters formed and draw inferences. Refer to crime_data.csv dataset.

	X	Murder	Assault	UrbanPop	Rape
1	Alabama	13.2	236	58	21.2
2	Alaska	10.0	263	48	44.5
3	Arizona	8.1	294	80	31.0
4	Arkansas	8.8	190	50	19.5
5	California	9.0	276	91	40.6
6	Colorado	7.9	204	78	38.7
7	Connecticut	3.3	110	77	11.1
8	Delaware	5.9	238	72	15.8
9	Florida	15.4	335	80	31.9
10	Georgia	17.4	211	60	25.8
11	Hawaii	5.3	46	83	20.2
12	Idaho	2.6	120	54	14.2
13	Illinois	10.4	249	83	24.0

3. Analyze the information given in the following 'Insurance Policy dataset' to create clusters of persons falling in the same type. Refer to Insurance Dataset.csv

	Premiums.Paid	Age	Days.to.Renew	Claims.made	Income
1	2800	26	233	3890.076	28000
2	2950	27	130	2294.444	29500
3	3100	28	144	2564.545	31000
4	3250	30	65	1978.261	32500
5	3400	32	56	2009.091	34000
6	3550	35	89	2349.455	35500
7	3700	44	95	2503.346	37000
8	3850	45	48	2217.405	38500
9	4000	46	76	2527.778	40000
10	6225	56	200	6908.232	41500
11	6450	67	211	7672.549	43000
12	6675	69	245	10208.824	44500
13	6900	70	261	12192.233	46000

4. Perform clustering analysis on the telecom dataset. The data is a mixture of both categorical and numerical data. It consists of the number of customers who churn. Derive insights and get possible information on factors that may affect the churn decision. Refer to Telco_customer_churn.xlsx dataset.

Customer ID	Count	Quarter	Referred a Friend	Number of Referrals	Tenure in Months	Offer	Phone Service	Avg Monthly Long Distance Charges	Multiple Lines	Internet Service	Internet Type	Avg Monthly GB Download	Online Security	Online Backup	Device Protection	Premium Tecl	Streaming TV	Streaming Movies	Streaming Music
8779-QRDMV	1	Q3	No	0	1	None	No	0	No	Yes	DSL	8	No	No	Yes	No	No	Yes	No
7495-OOKFY	1	Q3	Yes	1	8	Offer E	Yes	48.85	Yes	Yes	Fiber Optic	17	No	Yes	No	No	No	No	No
1658-BYGOY	1	Q3	No	0	18	Offer D	Yes	11.33	Yes	Yes	Fiber Optic	52	No	No	No	No	Yes	Yes	Yes
4598-XLKNJ	1	Q3	Yes	1	25	Offer C	Yes	19.76	No	Yes	Fiber Optic	12	No	Yes	Yes	No	Yes	Yes	No
4846-WHAFZ	1	Q3	Yes	1	37	Offer C	Yes	6.33	Yes	Yes	Fiber Optic	14	No	No	No	No	No	No	No
4412-YLTKF	1	Q3	No	0	27	Offer C	Yes	3.33	Yes	Yes	Fiber Optic	18	No	No	Yes	No	No	No	No
0390-DCFDQ	1	Q3	Yes	1	1	Offer E	Yes	15.28	No	Yes	Fiber Optic	30	No	No	No	No	No	No	No
3445-HXXGF	1	Q3	Yes	6	58	Offer B	No	0	No	Yes	DSL	24	No	Yes	Yes	No	No	Yes	No
2656-FMOKZ	1	Q3	No	0	15	Offer D	Yes	44.07	Yes	Yes	Fiber Optic	19	No	No	No	No	No	No	No
2070-FNEXE	1	Q3	No	0	7	Offer E	Yes	26.95	No	Yes	Fiber Optic	18	Yes	No	No	No	No	No	No

5. Perform clustering on mixed data. Convert the categorical variables to numeric by using dummies or label encoding and perform normalization techniques. The dataset has the details of customers related to their auto insurance. Refer to Autoinsurance.csv dataset.

Customer	State	Customer	Response	Coverage	Education	Effective To Date	Employme	Gender	Income	Location C	Marital Sta	Monthly Pi	Months Sir	Months Sir	Number o	Number o	Policy Typ	Policy	Renew Off	Sales Char	Total Clair	Vehicle Cla	Vehicle Siz
BU79786	Washingtc	2763.519	No	Basic	Bachelor	2/24/2011	Employed	F	56274	Suburban	Married	69	32	5	0	1	Corporate	Corporate	Offer1	Agent	384.8111	Two-Door	Medsize
QZ44356	Arizona	6979.536	No	Extended	Bachelor	1/31/2011	Unemploy	F	0	Suburban	Single	94	13	42	0	8	Personal A	Personal L	Offer3	Agent	1131.465	Four-Door	Medsize
AI49188	Nevada	12887.43	No	Premium	Bachelor	2/19/2011	Employed	F	48767	Suburban	Married	108	18	38	0	2	Personal A	Personal L	Offer1	Agent	566.4722	Two-Door	Medsize
WW63253	California	7645.862	No	Basic	Bachelor	1/20/2011	Unemploy	M	0	Suburban	Married	106	18	65	0	7	Corporate	Corporate	Offer1	Call Cente	529.8813	SUV	Medsize
HB64268	Washingtc	2813.693	No	Basic	Bachelor	3/2/2011	Employed	M	43836	Rural	Single	73	12	44	0	1	Personal A	Personal L	Offer1	Agent	138.1309	Four-Door	Medsize
OC83172	Oregon	8256.298	Yes	Basic	Bachelor	1/25/2011	Employed	F	62902	Rural	Married	69	14	94	0	2	Personal A	Personal L	Offer2	Web	159.383	Two-Door	Medsize
XZ87318	Oregon	5380.899	Yes	Basic	College	2/24/2011	Employed	F	55350	Suburban	Married	67	0	13	0	9	Corporate	Corporate	Offer1	Agent	321.6	Four-Door	Medsize
CF85061	Arizona	7216.1	No	Premium	Master	1/18/2011	Unemploy	M	0	Urban	Single	101	0	68	0	4	Corporate	Corporate	Offer1	Agent	363.0297	Four-Door	Medsize
DY87989	Oregon	24127.5	Yes	Basic	Bachelor	1/26/2011	Medical Le	M	14072	Suburban	Divorced	71	13	3	0	2	Corporate	Corporate	Offer1	Agent	511.2	Four-Door	Medsize
BQ94931	Oregon	7388.178	No	Extended	College	2/17/2011	Employed	F	28812	Urban	Married	93	17	7	0	8	Special Au	Special L2	Offer2	Branch	425.5278	Four-Door	Medsize
SX51350	California	4738.992	No	Basic	College	2/21/2011	Unemploy	M	0	Suburban	Single	67	23	5	0	3	Personal A	Personal L	Offer1	Agent	482.4	Four-Door	Small
VQ65197	California	8197.197	No	Basic	College	6/1/2011	Unemploy	F	0	Suburban	Married	110	27	87	0	3	Personal A	Personal L	Offer2	Agent	528	SUV	Medsize
DP39365	California	8798.797	No	Premium	Master	6/2/2011	Employed	M	77026	Urban	Married	110	9	82	2	3	Corporate	Corporate	Offer2	Agent	472.0297	Four-Door	Medsize
SJ95423	Arizona	8819.019	Yes	Basic	High Schoo	10/1/2011	Employed	M	99845	Suburban	Married	110	23	25	1	8	Corporate	Corporate	Offer2	Branch	528	SUV	Medsize

Hints:

1. Business Problem

- **1.1. What is the business objective?**
- **1.2. Are there any constraints?**

2. Work on each feature of the dataset to create a data dictionary as displayed in the below image:

Name of Feature	Description	Type	Relevance
ID	Customer ID	Quantitative, Nominal	Irrelevant, ID does not provide useful information

3. Data Pre-processing

- **3.1 Data Cleaning, Feature Engineering, etc.**

4. Exploratory Data Analysis (EDA):

- **4.1. Summary.**
- **4.2. Univariate analysis.**
- **4.3. Bivariate analysis.**

5. Model Building

- **5.1 Build the model on the scaled data (try multiple options).**
- **5.2 Perform K- means clustering and obtain optimum number of clusters using scree plot.**
- **5.3 Validate the clusters (try with different number of clusters) – label the clusters and derive insights (compare the results from multiple approaches).**

6. Write about the benefits/impact of the solution - in what way does the business (client) benefit from the solution provided?